HW 8 Part 3: Tools for 3D World Building and Spatial AI

Daniel Oni | CS 7382 Computer Graphics | Spring 2026

What I Looked At

I picked three things to compare across what I would call the new wave of 3D world tooling: Meta's SAM 3D, the Allen Institute's Objaverse and Objaverse-XL datasets, and NVIDIA's 3DGUT release on the Spatial AI side. They cover three different needs that come up if you are building a 3D scene now: getting an object out of a single photo, finding 3D assets to train on or drop into a scene, and rendering reconstructed real-world environments well.

Meta SAM 3D

Project page: https://ai.meta.com/research/sam3d/

SAM 3D came out in November 2025 alongside SAM 3, and it is two models. **SAM 3D Objects** reconstructs a 3D mesh and texture for an object selected in an image, using a two-stage diffusion transformer pipeline that first generates shape and pose, then refines texture. **SAM 3D Body** fits a parametric body model to a person in an image and is compatible with Meta's Momentum Human Rig, so the output is animatable. The really interesting feature is the Align pass that places multiple SAM 3D Objects outputs and SAM 3D Body outputs into a shared coordinate system, so a single photo can produce a populated mini-scene. There is a browser playground at the Segment Anything Playground if you do not want to run anything locally, and the code is open at https://github.com/facebookresearch/sam-3d-objects.

What hooked me about SAM 3D is the input requirement. Photogrammetry needs dozens of overlapping photos. NeRF and Gaussian Splatting need at least a video walkthrough. SAM 3D wants a single image. It can do that because it is a generative model that has seen enough 3D shape data during training that it can hallucinate the missing back side of an object plausibly. You give up some accuracy to gain a lot of convenience. For prototyping AR scenes or quickly placing assets, that tradeoff seems worth it.

Objaverse and Objaverse-XL

Project page: https://objaverse.allenai.org/

Objaverse is the dataset that explains where models like SAM 3D get their priors. **Objaverse 1.0** has over 800,000 annotated 3D objects with descriptive captions and tags. The follow-up, **Objaverse-XL**, is over 10 million 3D objects scraped from sources like GitHub, Sketchfab, Thingiverse, Polycam, and the Smithsonian. The paper is on arXiv (Deitke et al., 2307.05663), and the data is hosted on Hugging Face under *allenai/objaverse* and *allenai/objaverse-xl*. There is a Colab notebook on the project page for loading objects.

The size jump from "tens of thousands of curated 3D assets" (which is what older datasets like ShapeNet provided) to ten million scraped assets is the reason the current generation of 3D generative models works at all. It is the same scaling story that LAION told for 2D images, just a few years later for geometry. It also explains why outputs from SAM 3D and similar tools sometimes look generic. The training distribution is whatever people on the internet uploaded to Sketchfab, which is heavy on game props, anime characters, and 3D-printable trinkets, and lighter on serious architectural or industrial geometry.

NVIDIA Spatial AI: 3DGUT

Project page: https://research.nvidia.com/labs/toronto-ai/3DGUT/

This one bridges directly back to my Gaussian Splatting writeup. **3DGUT** is an NVIDIA Research output (CVPR 2025, Wu et al.) that swaps the EWA splatting formulation in standard 3D Gaussian Splatting

for an Unscented Transform borrowed from the Kalman filter literature. In practical terms, it lets Gaussian splats render correctly through nonlinear camera models like fisheye lenses, and it accommodates rolling-shutter effects from real phone cameras. It also aligns with their ray-tracing variant 3DGRT, so you can splat the primary ray and trace secondary rays for reflections and refractions in the same scene representation. The code is at https://github.com/nv-tlabs/3dgrut.

That last point is the one I find genuinely exciting. In the texture mapping unit we did, environment maps and reflections were faked using cubemaps that have to be authored or pre-baked. A representation that can render a real scanned environment and also do real reflections inside that environment would skip a lot of that authoring work.

If This Were Useful for Future Students

If a future class wanted to add a one-week unit on this stuff, the easiest path I would suggest is the Segment Anything Playground for SAM 3D (no install required) and a quick browse through Objaverse-XL on Hugging Face to look at what the dataset actually contains. The mental model that would carry forward from the rest of the class is: scenes can be represented as triangles (what we built), as fields (NeRF), or as splats (3DGS), and each representation determines what the renderer has to do. The new tools mostly differ in how they produce the representation, not in how the pixels actually get drawn at the end.

References and Where to Find Things

- Meta SAM 3D research page. https://ai.meta.com/research/sam3d/
- Meta SAM 3D announcement blog. https://ai.meta.com/blog/sam-3d/
- SAM 3D Objects code: https://github.com/facebookresearch/sam-3d-objects
- Objaverse project page. https://objaverse.allenai.org/
- Deitke et al., Objaverse-XL: A Universe of 10M+ 3D Objects, NeurIPS 2023, arXiv:2307.05663. https://arxiv.org/abs/2307.05663
- Hugging Face dataset: https://huggingface.co/datasets/allenai/objaverse-xl
- NVIDIA 3DGUT project. https://research.nvidia.com/labs/toronto-ai/3DGUT/
- NVIDIA Spatial AI SDK overview. https://research.nvidia.com/labs/sil/spatial-ai-sdks/
- 3DGUT/3DGRT GitHub. https://github.com/nv-tlabs/3dgrut